

High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks

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ABSTRACT

This paper develops an methodology to estimate safe separation distance for UAVs which share the same source of guidance and LADGNSS differential corrections.

1.0. INTRODUCTION

Recent technological improvements and increased military utilization have proven the operational viability of unmanned aerial vehicles (UAVs) and made them attractive for a wide range of potential civil applications. In particular, the capability and low cost of small UAVs have led to a dramatic growth in potential commercial applications over the past few years as well as a major interest in the media. While most current uses of UAVs are remotely piloted by humans, future UAVs will operate autonomously and coordinate their activities in large groups. Autonomous UAVs networks guided by central controllers will safely and routinely perform various missions such as surveillance, reconnaissance, and data collection [1].

Prior work [1, 2] has proposed a concept of UAV network operations that use differential corrections from Local-Area Differential GNSS (LADGNSS) to provide increased accuracy, safety, and reliable navigation to UAVs within 5 to 100 km of the controller station. LADGNSS supports low cost for commercial applications and high integrity to allow UAVs to operate in close proximity to each other while minimizing collision risk. Local-area UAV networks can perform their tasks safely as long as UAV locations within the network are

monitored by the controller to insure that safe separation distances are maintained from other vehicles and ground obstructions. The controller, including a LADGNSS reference station, receives GNSS signals and positions of UAVs, estimates errors and separation distances, and broadcasts differential corrections and optimized guidance for precise and safe operation of UAVs.

This paper presents a methodology to estimate the desired minimum separation distance that should be defined to prevent collisions by being utilized for guidance of UAVs. Separation distance can be derived by modeling mainly two sources of position uncertainty that could potentially lead to collisions: (i) Navigation System Error (NSE), and (ii) Flight Technical Error (FTE). This paper models these errors both theoretically and experimentally. Ionospheric and tropospheric models for NSE were developed theoretically. Though UAV flight experiments, we also determined airborne error model and FTE model. In this paper, Section 2.0 illustrates the concept of local area UAV network and component of safe separation distance between UAVs. Section 3.0 explains the methodology to estimate safe separation distance including the details of how to model the NSE and FTE. Section 4.0 shows flight test configuration we used for flight experiments. Section 5.0 shows the results of UAV flight test and simulation of safe separation distance between UAVs. Section 6.0 concludes the paper.

2.0. LOCAL AREA UAV NETWORK OVERVIEW

Figure 1 illustrates the concept of local-area UAV network operations developed in [1,2]. The control station, including a LADGNSS reference station, receives GNSS signals and positions of the UAVs. Then they estimate the local-area differential GNSS corrections and separation distances between UAVs. Finally, they broadcast

differential corrections and integrity information as well as real-time guidance based on safe separation distance for each UAV in the network. This allows the UAVs to operate in close proximity to each other with minimal risk of collision. The maximum operational range of a single network can be horizontally limited by datalink range more than the gradual degradation of LADGNSS differential correction. However, for the vertical area of operations, the UAVs fly in the space between the upper and lower limits, which define the flight space boundaries between the UAVs, manned aircrafts and ground obstacles.

In Figure 1, “in-network” UAVs refers to the UAVs within the same network which have the same source of guidance and LADGNSS differential corrections. “In-network” UAVs have both vertical and horizontal minimum separations between themselves for safe operations. “Out-of-network” UAVs represents all the other UAVs. They may share the use of GNSS, but do not share guidance or differential corrections. As they are assumed to fly in the same general airspace allocated to the UAVs, separations between “in-network” and “out-of-network” UAVs are also defined to avoid collisions with UAVs.

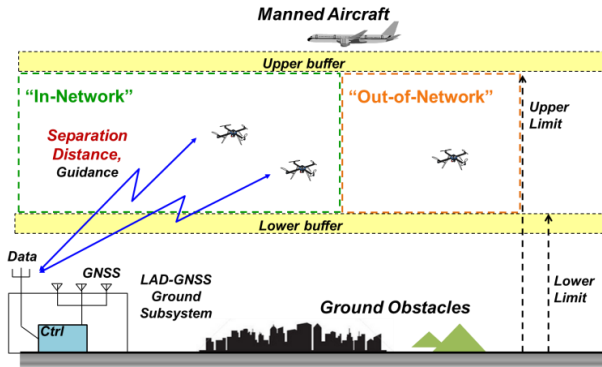


Figure 1

As shown in Figure 2, the separation distances defined above into mainly two error sources that could potentially lead to collisions. First is NSE, or the residual error in the estimated position from the true position which cannot be completely eliminated by LADGNSS. Second, FTE is error in following the path defined by the guidance source, which depends on the controller’s performance and external disturbances such as wind. In Figure 2, “Target” is the obstacle to be avoided and can represent another vehicle, including “in-network” and “out-of-network” UAVs. To get the safe separation distances, the error sources of each UAV and that of the targets are reasonably modeled. In the local-area UAV network, UAVs are commonly operated in the “in-network”. Therefore, we are interest in the separation distances between the UAVs in the “in-network”. As LADGNSS navigation errors for nearby “in-network” UAVs will be

very highly correlated, the only the uncorrelated error components contribute to the potential separation violations.

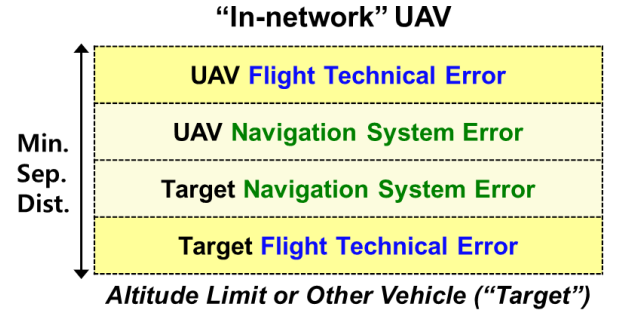


Figure 2

3.0. MODELING(?) METHODOLOGY

In local area network system, even after applying differential corrections, residual errors still remain. However, as the correlated errors are canceled out for all the UAVs in the same network, only the uncorrelated error components contribute to the potential separation violations. To estimate the uncorrelated errors, theoretical error models for the tropospheric and ionospheric errors were developed. We also determined error models which depend on hardware, environmental and operation conditions experimentally. These errors include the airborne multipath error, the ground facility errors, and the flight technical errors. However, as the in-networked UAVs share the same correction broadcasted from the ground facility, the correlated errors are canceled out although the ground facility errors can be different from those of the standard GBAS ground facility.

3.1. RESIDUAL TROPOSPHERIC ERROR MODELING

As each UAV is at a different altitude than the ground reference stations, a differential tropospheric delay correction is required. For Ground Based Augmentation System (GBAS) applications, the vertical tropospheric correction in meters for a user at altitude Δh meters above the reference station can be given by

$$TC_{vert} = 10^{-6} N_R h_0 (1 - e^{-\Delta h / h_0})$$

with

N_R = refractivity index transmitted by ground subsystem [].

h_0 = tropospheric scale height transmitted by ground subsystem (in meters) [].

For a satellite at elevation angle θ , the slant (i.e., along the actual path between satellite and receiver) tropospheric correction is defined as follows:

$$TC = \frac{N_R h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h/h_0})$$

Using the refractivity uncertainty transmitted by ground subsystem (σ_N), the residual tropospheric error bound for GBAS is also defined by

$$\sigma_{tropo} = \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h/h_0})$$

For the LADGNSS errors among in-network UAVs, we are interested in the differential tropospheric delay as observed at an UAV_a Δh_a meters above the reference station and at another UAV_b Δh_b meters above the reference station. As the correlated tropospheric error between two UAVs is canceled out, the residual tropospheric error bound between UAVs is computed as

$$\begin{aligned} \sigma_{tropo, between UAVs} &= \sigma_{tropo}(UAV_b) - \sigma_{tropo}(UAV_a) \\ &= \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (e^{-\Delta h_b/h_0} - e^{-\Delta h_a/h_0}) \end{aligned}$$

3.2. RESIDUAL IONOSPHERIC ERROR MODLEING

Ionospheric temporal and spatial decorrelation can lead to residual range errors to a LADGNSS user. The temporal decorrelation refers to the ionosphere pseudorange delay changes over time at the same point in the ionosphere. If users use same carrier-smoothing time constants with reference station, the temporal decorrelation does not lead to steady state residual range error. As we assume that carrier-smoothing time constants are matched between reference station and in-network UAVs, the spatial decorrelation, where the delay changes from point to point in the ionosphere at a given time, is only interested.

The residual ionospheric spatial decorrelation error model for UAVs is an extension of the original work, which derived a simplified model (σ_{iono}) of residual user range error due to spatial decorrelation for GBAS. The original model was for ionospheric decorrelation errors between a fixed reference location and approaching aircraft. Therefore, in this paper, a generalized version of the ionospheric model which considers both UAVs as moving was developed. The differential carrier-smoothed ionosphere delay between UAV_a and UAV_b at epoch k ($\Delta \hat{I}_k$) can be written as:

$$\Delta \hat{I}_k = \frac{\partial I_r}{\partial x_r} x_r - 2(\tau - T_s) \frac{\partial I_b}{\partial x_b} v_{b,k-1} + 2(\tau - T_s) \frac{\partial I_a}{\partial x_a} v_{a,k-1}$$

where x_r is range distance (in meters) between UAV_a and UAV_b ; $\partial I_r / \partial x_r$ is the spatial ionospheric gradient between UAVs; τ is the carrier smoothing time constant (in seconds); T_s is the sampling time (in seconds); $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ are the spatial ionospheric gradients in the direction of motion of the UAV_a and UAV_b respectively; and v_a and v_b are the velocity of the UAV_a and UAV_b respectively. This differential ionospheric delay consists of the spatial ionospheric decorrelation between the UAVs, which express as $\partial I_r / \partial x_r \cdot x_r$, and the smoothing effects. As smoothing effects are functions of the velocity of the UAVs and the spatial ionospheric gradients in the direction of motion of the UAVs, ionospheric model for UAVs has been more complex when we consider both UAVs as moving.

To derive the model, two conditions are assumed. First, ionosphere is a spatially linear ionospheric decorrelation with gradient of G as shown in Figure 3. In the Figure 3, α represents an angle between ionospheric front with gradient of G and line of sight between UAVs. The θ_a and θ_b are angles between line of sight between UAVs and a vector of velocity of the UAV_a and UAV_b respectively. In this assumption, $\partial I_r / \partial x_r$, $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ can be obtained as

$$\frac{\partial I_r}{\partial x_r} = G \cos \alpha$$

$$\frac{\partial I_a}{\partial x_a} = G \cos(\alpha + \theta_a) = G(\cos \alpha \cos \theta_a - \sin \alpha \sin \theta_a)$$

$$\frac{\partial I_b}{\partial x_b} = G \cos(\alpha + \theta_b) = G(\cos \alpha \cos \theta_b - \sin \alpha \sin \theta_b)$$

Therefore, the differential carrier-smoothed ionosphere delay can be formulated as

$$\begin{aligned} \Delta \hat{I}_k &= \\ &G \left[\cos \alpha (x_r - 2(\tau - T_s) v_{b,k-1} \cos \theta_b + 2(\tau - T_s) v_{a,k-1} \cos \theta_a) \right. \\ &\quad \left. + \sin \alpha (2(\tau - T_s) v_{b,k-1} \sin \theta_b - 2(\tau - T_s) v_{a,k-1} \sin \theta_a) \right] \end{aligned}$$

Using the trigonometric function, the differential delay can be simply expressed as

$$\begin{aligned} \Delta \hat{I}_k &= A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \sin(\alpha + \varphi), \\ \sin \varphi &= \frac{A}{\sqrt{A^2 + B^2}} \end{aligned}$$

To conservatively bound the errors, we assumed second condition, which the ionospheric slope is located in the

direction that maximizes the differential delay between UAVs. On the other hands, we determine the $\sin(\alpha + \varphi)$ in Eq.7 as 1.0. And then, using one-sigma value for nominal ionosphere vertical spatial gradient (σ_{vig}), the residual ionospheric error bound between UAVs is derived as

$$\sigma_{iono,betweenUAVs} = F_{pp} \times \sigma_{vig} \sqrt{\left(x_r - 2(\tau - T_s)v_{b,k-1} \cos \theta_b + 2(\tau - T_s)v_{a,k-1} \cos \theta_a\right)^2 + \left(2(\tau - T_s)v_{b,k-1} \sin \theta_b - 2(\tau - T_s)v_{a,k-1} \sin \theta_a\right)^2}$$

where F_{pp} is the vertical-to-slant obliquity factor for a given satellite.

$$F_{pp} = \left[1 - \left(\frac{R_e \cos \theta}{R_e + h_l}\right)^2\right]^{-\frac{1}{2}}$$

with

R_e = radius of the earth = 6378.1363 km.

h_l = ionospheric shell height = 350 km

θ = the elevation angle of satellite

If this model applies the ionospheric decorrelations between a fixed reference location and approaching aircraft which are considered for development of original model, which v_a is zero, v_b is v_{air} , θ_a is 0° , and θ_b is 180° , we can obtain the formula as following.

$$\sigma_{iono} = F_{pp} \times \sigma_{vig} \times (x_r + 2 \times (\tau - T_s) \times v_{air})$$

This equation is same with the ionospheric model for GBAS. This case study indicates that the proposed ionospheric model is a generalized version of residual ionospheric error model encompassing the original model.

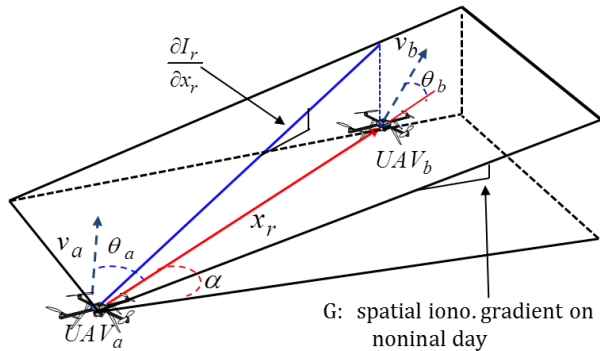


Figure 3

3.3. AIRBORNE MULTIPATH MODLEING

As the UAVs fly relatively close to ground objects, significant airborne multipath can be presented from ground reflections. The airborne multipath errors also are influenced by the entire airborne implementation including the antenna, receiver, and the smoothing. Therefore, the airborne error is modeled based on UAV flight experiments. To estimate multipath errors, the dual frequency GPS data were collected by the UAVs and analyzed. The overall data analysis process is illustrated in Figure 3.

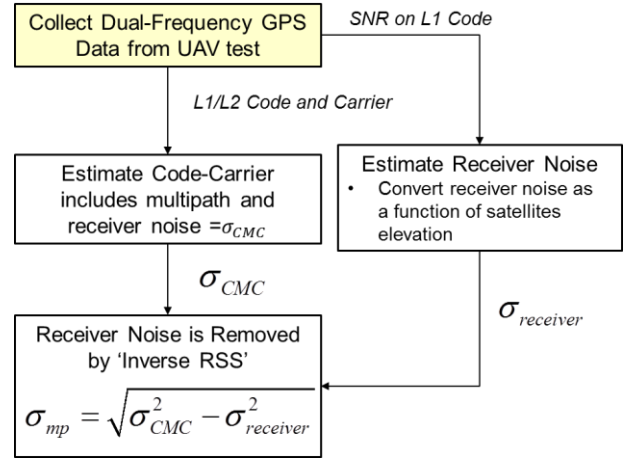


Figure 4

The dual frequency GPS data has been analyzed by differencing the code and carrier phase observables for each satellite as shown Eq.7.

$$\rho_{L1} - \phi_{L1} = 2I + M_{L1} - m_{L1} - N_{L1} + \varepsilon_\rho + \varepsilon_\phi$$

Differencing the code and carrier phase observables leaves only the errors due to code and carrier multipath (M_{L1} and m_{L1}), errors due to receiver noise on code and carrier measurement (ε_ρ and ε_ϕ), errors due to the divergence between the code and carrier phase as the signal passes through the ionosphere (I), and integer ambiguity (N_{L1}). In this equation, carrier phase multipath and noise errors are considered to be negligible and the integer ambiguity can be considered a removable bias as long as there are no cycle slips. Finally, if the effect of the ionospheric divergence is corrected by using the dual frequency carrier phase observables, the final result is an estimate of a combination of both multipath and receiver noise variations for each satellite as shown Eq.7.

$$I = \frac{\phi_{L1} - \phi_{L2}}{\gamma - 1}$$

$$\rho_{L1} - \phi_{L1} - 2 \frac{\phi_{L1} - \phi_{L2}}{\gamma - 1} = M_{L1} + \varepsilon_{\rho}$$

If the multipath and receiver noise are independent and all other error sources are effectively removed from the CMC observations, then the thermal noise is removed by inverse RSS (Root Sum Square):

$$\sigma_{mp} = \sqrt{\sigma_{CMC}^2 - \sigma_{noise}^2}$$

To separate out receiver noise portion, we computed theoretical receiver noise as shown Eq.7 and subtracted it from code-carrier measurements.

$$\sigma_{noise} = \lambda_c \sqrt{\frac{d}{8C / N_0 \tau}}$$

where λ_c is code chip wavelength (in meters), d is correlate spacing, C/N_0 is carrier to noise value in units of ratio-Hz, and τ is the smoothing time constant.

3.4. FLIGHT TECHNICAL ERROR MODLEING

In order to estimate flight technical error (FTE), firstly, it is required to set-up path-following guidance for UAV. In this research, flight path is designed to cover interesting area considering practical applications of UAV such as agriculture, surveying, mapping, and reconnaissance. Figure (?) shows designed flight path. The path consists of three straight sections and two semicircular sections. The ground altitude of each waypoint is fixed as 10m.



Figure 5

UAV flight test is conducted along the path, and flight trajectory information is logged at 10 hertz during the test. ??

Lateral and vertical FTE are calculated by subtracting designed guidance from logged position in each direction. However, the entire FTE data cannot be used to estimate statistical characteristics, because assumption that each error samples are independent is invalid. Consequently, use of the entire error samples can causes significant autocorrelation in the FTE data. Therefore, resampling FTE points along flight path with constant distance interval is required [10]. First, in order to evaluate the significance of autocorrelation at k^{th} lag for N samples, the autocorrelation function is defined as follows:

$$\gamma_k = \frac{\sum_{i=1}^{N-k} (x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

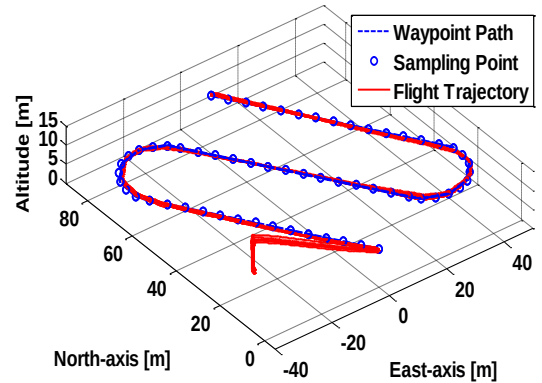
where x_i is i^{th} data sample and $\bar{x}$ is mean of data samples. The significance of autocorrelation at the k^{th} lag is tested to determine the distance interval between samples. The null hypothesis that k^{th} lag auto correlation value is statistically zero can be expressed as:

$$H_0 : \rho_k = 0$$

The hypothesis is evaluated with the critical values for type I error level α .

$$\gamma_{\alpha,k} = N^{-1}(\alpha, 0, 1) \sqrt{\frac{1}{n} \left(1 + 2 \sum_{i=1}^k \gamma_i^2 \right)}$$

where $k = 1, \dots, \text{int}(N/4)$ and $N^{-1}(\alpha, 0, 1)$ is standard normal variate of standard normal inverse CDF at α . The null hypothesis at k^{th} lag is rejected if $\gamma_k > \gamma_{k,\alpha}$.



3.5. SAFETY RISK ASSESSMENT

The K-factor, which is multiplied to the error models, is obtained based on the risk assessment methodology in [2]. The probability that an "in-network" UAV collides with other vehicles and thereby causes harm to human is the key to determining how much error must be allowed for. A collision of UAV in the air has the possibility of falling to the ground. The wreckage dropped may lead to harm to nearby people. Therefore, the probability of harm to humans (state "HH"), given a collision (state "CO") resulting from violation of separation standards (state "SV") can be defined as:

$$\Pr(HH | SV) = \Pr(HH | CO) \times \Pr(CO | SV)$$

Table 1

Scenario	Pr(CO SV) (mean)	Pr(CO SV) (mean)	Pr(CO SV) (mean)	Pr(CO SV) (worst case)
In-network UAV	10 ⁻⁴	10 ⁻⁵	10 ⁻⁹	10 ⁻⁵

The Table 1 that shows the probabilities of collisions leading to harm to humans given violation of the minimum separation was proposed in [1]. The "worst case" probability assumed that the probability of collision given a violation of separation is 1.0. Although the mean probability of collision given a violation of separation is very small, the traditional means of assuring safety under anomalous conditions may require the assumption of "worst-case" conditions rather than average ones. Using this risk model with "worst-case" condition, the probability of the violation of separation is determined when the estimate of the probability of harm to humans is given. We can borrow 10⁻⁹ that is the smallest probability required by the safety probability in the FAA hazard risk index [1]. If 10⁻⁹ is the highest allowed risk of harm to humans that can be tolerated from violation of UAV separation, given in Eq. 2, the probability of violation of separation is determined as 10⁻⁴.

$$\Pr(HH | SV) \times \Pr(SV) \leq 10^{-9}$$

Therefore the Gaussian K-factor of 3.89 is used to represent a double-sided tail probability of 10⁻⁴.

4.0. FLIGHT TEST CONFIGURATION



UAV Specification	
Platform	Octo-rotor
Diameter	110 cm
Height	35 cm
Weight	5.31 kg
Flight Time	10 min
Battery	LiPo 6s 22.2V, 5000mAh
Payloads	
GPS Antenna	NovAtel OEM-V3
GPS Receiver	NovAtel ???
Flight Controller	Ardupilot Mega 2.6
Modem for DGPS	Xbee Pro Series 1

5.0. RESULT

5.1. AIRBORNE MULTIPATH MODELING

Throughout the UAV flight test, the data was collected from single receiver using 0.1 chip correlator spacing and types as described in section 4.0. A total of 49 flight test was conducted, which corresponds to the total flight time of 4 hours 3minutes. In these flights, UAVs hovered at the flight altitude of 40 m, which is a lot lower than that of manned aircrafts, but still in the flight region of UAVs. Therefore the errors most likely contain multipath errors from ground reflection. However we couldn't confirm it yet. The multipath estimates are sorted onto 5° elevation bins and the standard deviation is computed. Figure 3 shows the standard deviation of multipath errors for the data from all UAV flight tests compared to the standard multipath airborne multipath model proposed in [3]. The standard airborne multipath model is given by:

$$\sigma_{Std.multipath}(\theta) = 0.13 + 0.53e^{(-\theta/10^\circ)}$$

The data were reasonably well-bounded by the standard multipath model shown as the red curve. Therefore, it is reasonable that the standard multipath error model is used for bounding the multipath errors of UAVs.

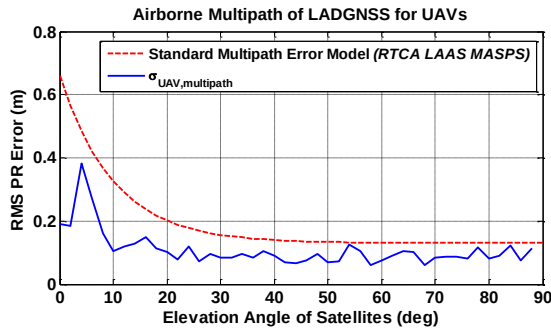


Figure 6

5.2. FLIGHT TECHNICAL ERROR MODLEING

Total 21 of path-following flight are conducted, and 51236 of FTE samples are collected. Following the resampling process noted in preceding section, sampling distance is determined as 5.15m when type I error level α is 99.5%. As a result of the resampling, only 1239 of FTE samples are left.

FTE error data are categorized into FTE in straight segment and FTE in curved segment. Figure 6 shows that lateral FTE RMS error in curved segment is much greater than other FTE errors. This is because the lateral FTE in curved segment is directly influenced by the momentum of the UAV. In addition, the UAV maneuvers on the horizontal plane more frequently when it makes turn around than when it just go straight, so it can also increase lateral FTE. On the other hand, the difference between the vertical FTEs of straight segment and curved segment is very small. This is because there is no variation in vertical guidance.

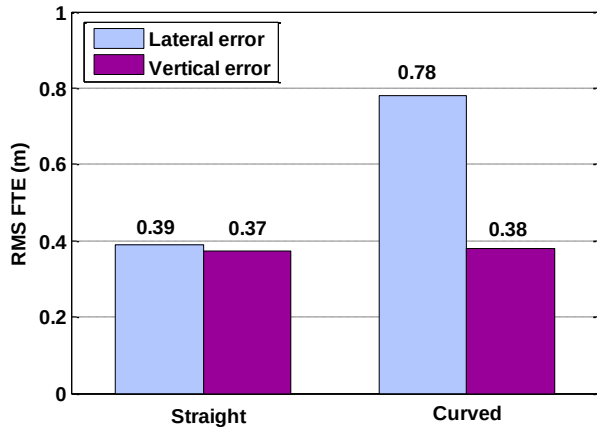


Figure 7

5.3. SEPARATION DISTANCE

From the all error models that estimated above sections and K-factor, simulations of safe separation distances

between in-network UAVs were conducted. For this simulation, the vertical distance between UAVs was fixed as 20 meters and the horizontal distance between UAVs as 30 meters. RTCA 24 constellation was used for satellite constellation. For FTE, straight flight mode was used. The Figure 7 shows the result of simulation of safe separation distances. In this figure, the horizontal separation distance, which is shown as red curve, is mostly below 3 meters. For vertical separation distance, it's about 3 to 5 meters and max at about 6 meters. The variation in time is coming from the GPS constellation change.

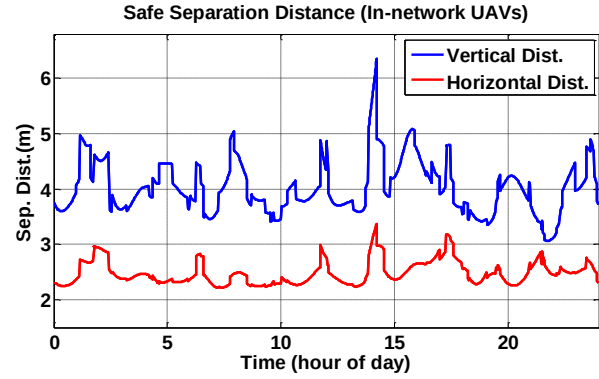


Figure 8

6.0. CONCLUSION

ACKNOWLEDGMENTS

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[1]